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Investigating the application of a Recommender System for Irish college courses

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Masters in Computer Science

Supervisor: Owen Conlan

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Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

Brendan McCann

March 12, 2026

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...ABSTRACT... **TODO**

Acknowledgments

TODO

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[3] `https://www.simplypsychology.org/herzbergs-two-factor-theory.html`

Chapter 1

Introduction

1.1 Motivation

This section describes the context behind known problems in the research domain and provides some background knowledge. In light of these established problems, a technology is introduced, one that could be appropriately applied to this domain. The applicability of the proposed solution is also discussed, all relevant issues in describing the research motivation behind this dissertation.

According to Higher Education Authority (2024); Department of Education (2024a), 79% of Irish secondary school students progressed to college after graduation. The individuals who have considered progressing to college have all been posed with the question: *What college course should I study?* This is a daunting question for many reasons. The majority of secondary school students are faced with making this decision at 16-19 years old, an age with limited life experience and knowledge about the world. In Smyth et al. (2011), Irish students were reported to experience significantly increasing levels of stress in their final year of secondary school, complicating this decision further. This decision also comes with many personally confronting questions, which may include: *"What do I want to do with the rest of my life?"* and *"What do I love doing?"*, among many others. Moreover, there exists the fear of choosing a college course the individual regrets. For instance, in Higher Education Authority (2018), 31% of recent Irish college graduates reported they were not likely to study their college course again, and in Roese and Summerville (2005), vocational mismatch is said to be a frequent regret for many Americans, adding more weight to this crucial life decision.

Choosing a college course is not only a challenging decision to wrestle with as an individual. Individuals are shown to be influenced by a multitude of external factors when choosing their college course. For instance in Aydin (2015); Leighton and Speer (2023), parents are shown to be one of the most influential factors of an individual's college course choice. Aydin (2015); Dickson (2010) have also shown that an individual's race/ethnicity, and their parent's educational and financial background are among the most prominent societal influences of an individual's college course choice, adding complexity to a decision that is already challenging.

To assist students in this difficult decision, it is worth introducing the role of a Guidance Counsellor (GC). GC's are trained professional in Irish secondary schools, primarily known for assisting students in their career prospects after graduation. It is worth noting that the role of a GC extends well beyond vocational support - according to Institute of Guidance Counsellors (2016), GC's are also trained to provide educational, personal and social support. In this context, it is worth introducing the Institute of Guidance Counsellors (IGC) [1], the professional body for GC's in Ireland. In Institute of Guidance Counsellors (2022), mental health issues were more commonly encountered by GC's compared to career decision-making issues [2], marking the first time that career decision-making issues were not the most commonly encountered issue by GC's since 2019. In the IGC's 2024-2027 strategic plan outlined in Institute of Guidance Counsellors (2024b), the IGC noted this stronger prevalence of mental health issues compared to career decision-making issues, reflecting their change in strategy for GC's going forward, signalling a shift in priorities for GC's across Irish secondary schools: vocational support may not be the most important aspect of a GC's role going forward.

We can shift our focus to the more technical side of things. A Recommender System (RS) is a technology that recommend items to a user. Well-known examples of RS's include Netflix [3] recommending films to their customers, or Amazon [4] suggesting products to their customers. One of the earliest known RS's was *Tapestry* in Goldberg et al. (1992), a system that made tailored mailing list recommendations to its users. Around this time, the field of RS's received considerable interest as e-commerce websites could increase their revenue if they suggested more personalised products to their users. As discussed in Schafer et al. (1999), Amazon and Netflix are the most notable adopters of RS's in order to increase company revenue.

[1] `https://igc.ie/`

[2] In Institute of Guidance Counsellors (2024a), career decision-making issues did surpass mental health issues as the most commonly encountered issue for GC's, however, both are pressing issues.

[3] `https://www.netflix.com/`

[4] `https://www.amazon.com/`

As previously explained, an RS recommends *items* to a *user*. However, we can reconceptualize an RS in the context of college courses: *college courses* can take the place of *items*, and a *student* can take the place of a *user*. Given the established difficulty for an individual choosing a suitable college course - including internal preferences and external influences, the shifting priorities of GC's towards student's mental health as opposed to career-making decisions, and a technology that could be appropriately applied to this domain, an exciting opportunity presents itself: A Recommender System for Irish College Courses.

A technology, such as an RS, could aid an individual in navigating a difficult decision-making process. For instance, there are over 45 higher education institutes in Ireland offering a total of over 1700 courses [5], representing a huge set of choice to the individual. An advantage of a personalised RS is to provide a user with a much smaller list of relevant recommendations. Long et al. (2025) demonstrated that smaller recommendation sets are often equally as effective as larger recommendations sets, suggesting that a much narrower set of personalised college course recommendations could be effective in this space.

For the 2026/2027 academic year, 52 [6] new higher education courses were introduced in Ireland. Moreover, as discussed in Section 2.4.1, the requirements for Irish higher education courses change every year as a reflection of demand and changing priorities. This introduces administrative strain on GC's, making it difficult to have up-to-date awareness on the newest courses and current course entry requirements. A technology, such as an RS, can be automated to hold the latest college course information and utilise this information in recommending college courses that a human GC may not even be aware of, posing a potential advantage of an RS in this domain.

In this section, we established the numerous difficulties posed to an individual in choosing a suitable college course. These difficulties include the sheer weight of this decision, and the external, societal factors that influence an individual's college course decision. We then discussed the shifting priorities of GC's from vocational to personal support, and introduced a technology which could be applicable to this domain - a Recommender System. A discussion was given into the ways in which a college course RS could alleviate this problem for both students and GC's, ultimately describing the motivation behind this research.

[5] Higher education colleges and courses can be found here: `https://careersportal.ie/`

[6] See `https://careersportal.ie/courses/coursefinder?types_in=2` for new courses

1.2 Research Question

In light of the motivation behind this research, the Research Question for this dissertation can be formulated: *How can a Recommender System assist an Irish student in their college course decision-making process?*

As we established Section 1.1, choosing a college course is a difficult, multi-faceted decision. In this dissertation's Research Question, "assist" simply refers to helping the individual navigate the difficult process of choosing a college course. A more elaborate discussion of how a college course RS can assist an individual in a process like this is given in **TODO**.

1.3 Research Objectives

With the motivation and research question formulated, the research objectives for this dissertation are outlined below:

1. Conduct thorough background research in answering the question: *"What college course should people study?"*
2. Identify the relevant information to question a user in order to make suitable college course recommendations.
3. Create an Irish college course RS with **TODO** statistically significant improvements compared to a baseline college course RS.
4. Analyse user evaluation metrics gathered from a conducted experiment.
5. Explore the qualities that would make an effective college course RS.

1.4 Overview

In this section, an overview of this dissertation's chapters are provided.

- Chapter 2- Background Reading. This chapter describes the necessary background knowledge to understand the Irish secondary school and college admission system.
- Chapter 3 - Literature Review. This chapter provides the reader with a solid understanding of the nuances behind an individual's college course choice. There will be explorations into an individual's motivations behind choosing a college course, the

external influences of an individual's college course choice, and the relevant information to question a user in order to make suitable college course recommendations, among other relevant discussions. As well as this, there will be an exploration into existing implementations of college course RS's, both in literature and real-world products. This includes a discussion into their successes and limitations.

- Chapter 4 - Initial Prototypes. **TODO**.
- Chapter 5 - Design. **TODO**
- Chapter 6 - Implementation. **TODO**
- Chapter 7 - Methodology. This chapter describes the experiment methodology in which the RS was tested with human participants.
- Chapter 8 - Evaluation. This chapter provides an objective analysis of the experiment evaluation metrics, and provides a qualitative analysis of verbal feedback gathered from the experiment.
- Chapter 9 - Conclusions & Future Work. **TODO**

Chapter 2

Background Reading

As mentioned in Chapter 1, this dissertation specifically focuses on a college course RS in the context of the Irish secondary school and college system. Grading systems and college admission systems often differ between countries, so this chapter seeks to provide essential background knowledge behind the Irish secondary school and the college admission system. There will be an explanation of the Leaving Certificate - the final Irish secondary school exam - the National Framework of Qualifications, a categorisation of Irish college courses, and the Central Applications Office, the primary college admission system in Ireland. As well as this, there will be a description of Irish college entry requirements, the fees and scholarships of undergraduate Irish college courses, and a greater explanation of the role of a GC in Irish secondary schools. Upon completion of the chapter, the reader should be familiar with the workings of the Irish secondary school and college admission system.

2.1 The Leaving Certificate

The Leaving Certificate (LC) is the final examination taken by the majority of secondary school students in Ireland. Instead of taking the LC, students can take the Leaving Certificate Applied (LCA) which is a more practical alternative to the LC. Students who take the LCA cannot directly enter higher education institutes; they commonly go for direct employment, apprenticeships or enter further education institutes. For the sake of narrowing scope, apprenticeships and further education courses are considered beyond the scope of this RS. The RS proposed in this dissertation will only recommend higher education college courses in Ireland.

A student's overall performance in the LC is measured in terms of points, a number that ranges between 0 and 625, where 625 is the maximum achievable points. A student chooses to be examined in each subject at increasing levels of difficulty, known as Higher Level (HL), Ordinary Level (OL) or Foundation Level (FL) [1]. A student receives a grade depending on the level and mark they achieve in the subject. For example, a student that gets a mark of 91% in the HL LC English exam receives a H1 grade, for which they receive 100 points, and a student that gets a mark of 64% in their OL LC Geography exam receives an O4 grade, for which they receive 28 points. For a full list of LC exam grades and their translated LC points, see Figure 2.1 below. A student is examined in a minimum of 6 subjects, and their overall LC points is an accumulation of the points received in each examined subject.

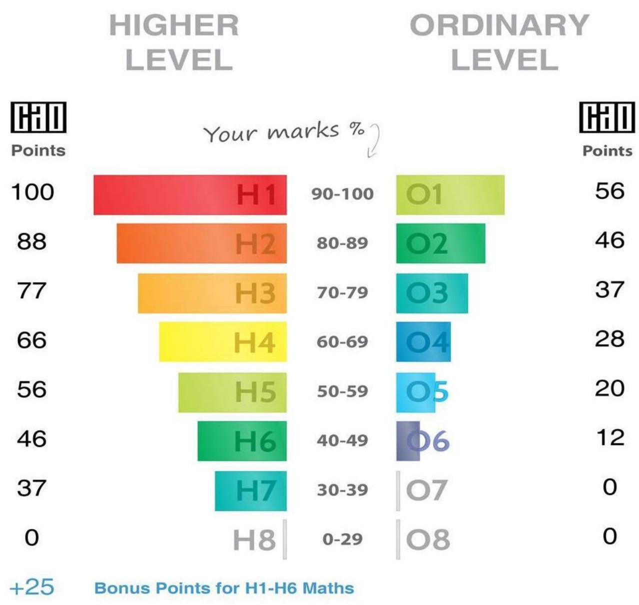

Figure 2.1: A full list of LC exam grades and their translated LC points.

2.2 NFQ Levels

The Irish National Framework of Qualifications (NFQ) [2] is a system that describes all education qualifications in Ireland. It is a 10-level system, where 1 is the lowest and 10 is the highest (doctorate degree). For instance, the LC is classed as NFQ Level 5. The NFQ levels that are relevant to this dissertation are NFQ Level 6 (Advanced or Higher Certificate), NFQ Level 7 (Ordinary Bachelors degree), NFQ Level 8 (Honours Bachelor degree) and NFQ Level 9 (Masters degree). A full graphic of the 10 NFQ levels are shown below in figure 2.2.

[1] Foundation Level is only offered for the LC Irish and Mathematics subjects.
[2] https://www.qqi.ie/what-we-do/the-qualifications-system/national-framework-of-qualifications

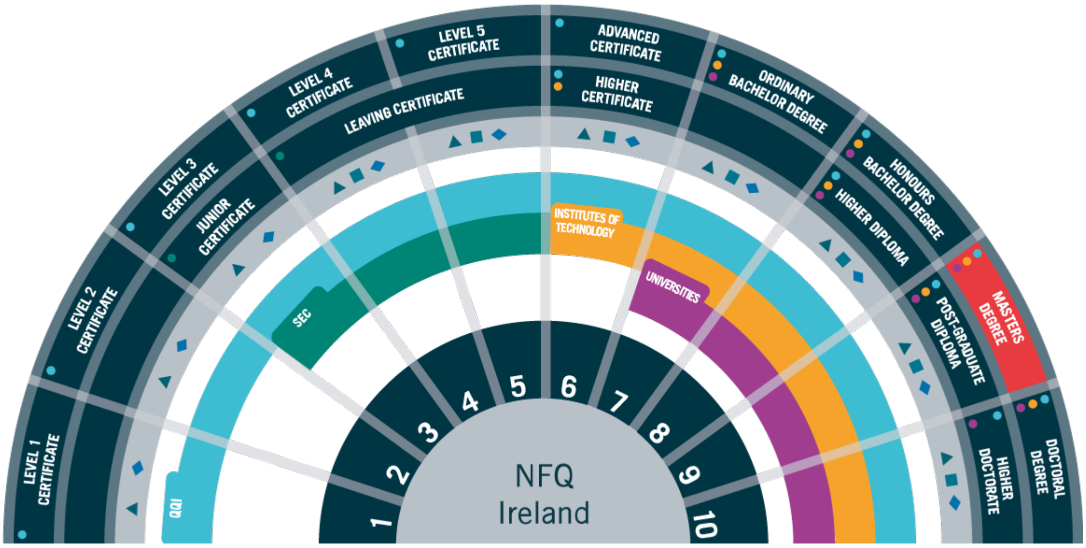

Figure 2.2: The 10-level NFQ system. Source: [3]

2.3 The Central Applications Office

Rather than going to the colleges directly to apply for a specific college course, in Ireland all college applications are done through the Central Applications Office (CAO) [4], the centralised college admissions system in Ireland. For reference, there were over 83,000 applications to the CAO in 2025 according to Central Applications Office (2025). As part of a CAO application, students are encouraged to create two numbered lists, each with up to 10 college courses. One of these numbered lists are for NFQ Level 8 college courses, and the other list is for NFQ Level 6 and 7 college courses. These college courses are ordered by the student's preference of study - for instance, the student's number one choice would be the course they want to study the most, and their number ten choice is the course they want to study the least out of this list.

The preference ranking of a student's CAO application courses is relevant here. For example, if a student receives an offer for a course that is ranked first in their list, they will not receive offers for courses below this preference, even if they satisfy the course's entry requirements. Entry requirements will be explained in section 2.4.

2.4 Entry Requirements

Entry requirements for colleges differ by country, and in this section the entry requirements for Irish college courses are described.

[4] https://www.cao.ie/

2.4.1 LC Points

The primary entry requirement for most college courses in Ireland is LC points. For example, the Computer Science (CS) course in Trinity College Dublin (TCD) has a minimum entry requirement of 533 points [5], meaning that a student must achieve 533 points or higher in their LC to be considered admission into the course. A course's LC points is the points of the student who was last admitted to the course in the previous academic year. For example, let's suppose that in 2025, 150 people were admitted into CS in TCD. Out of all these 150 students, the student with the lowest LC points was 533, meaning the course has a minimum entry requirement of 533 LC points. As a result, the LC points associated with a course are not fixed and fluctuate every year. LC points are commonly misunderstood as reflecting the difficulty of a course, however, it would be more accurate to say that the LC points reflect the *demand* of a course.

There are a handful of ways in which a student can achieve additional LC points. For example, A student will receive an extra 25 points if they achieve a passing ($\geq$40%) grade in the HL Mathematics exam. As well as this, if a student is takes their LC exams in the Irish language, they can receive up to 10% in bonus marks in each of their LC exams [6].

2.4.2 LC Subject Requirements

Many Irish college courses have grade requirements for specific subjects to allow admission into the course. In most cases, the subjects are relevant to the course and the college seeks knowledge in that subject. For example, CS in TCD has a minimum entry requirement of a H4 (60-69% in HL) in Mathematics to be considered for admission into the course.

In Ireland, students are obligated to take the LC English, Mathematics and Irish [7] exams, and they choose a minimum of four additional subjects to take exams in. The student makes their subject choices in fifth year, which is the penultimate year in Irish secondary schools. Due to the differing subject entry requirements by college courses, some students may be unable to take certain courses simply because they have not taken the relevant subject. For example, if a student has not taken two science [8] subjects, they are unable to be admitted into Medicine [9], meaning that a student's college course options

[5] 533 points as of 2025, see: https://www.tcd.ie/courses/undergraduate/courses/computer-science/

[6] for exact bonus marks in answering the LC in the Irish language, see: https://www.citizensinformation.ie/en/education/state-examinations/leaving-certificate/

[7] Some students may qualify for an exemption from Irish if they satisfy certain requirements.

[8] LC Physics, Chemistry, Physics/Chemistry, Biology and Agricultural Science.

[9] Subject requirements for TCD Medicine as of 2026: https://www.tcd.ie/courses/undergraduate/

are already limited by the subject choices they made in fifth year, almost two years before they finalise their CAO college course list.

In addition to specific college courses requiring a minimum grade in particular subjects, some subjects are required to be admitted into the college itself. For example, all Royal College of Surgeons in Ireland courses [10] require a third language [11].

2.4.3 Portfolios, Tests or Interviews

In a majority of cases, if a student satisfies the minimum entry requirements for a course with their LC points and subject grades, the student is admitted into the course. However, some Irish college courses have additional entry requirements. For instance, some courses may require the student to be examined in a test outside of the Leaving Certificate. For example, all Medicine courses in Ireland require their applicants to sit the Health Professions Admissions Test (HPAT) [12]. Some college courses may also require a portfolio submission, or for the student to undertake an interview. For example, the Architecture course in Technological University Dublin [13] requires the student to submit an additional portfolio showcasing the student's competence in sketching, painting, etc., as well as an interview demonstrating their interest in the course. Some courses may also require students to go through Garda Vetting [14], a background check for individuals working with children or vulnerable adults, an example of this includes General Nursing in TCD [15].

In the case of Portfolios and Tests, the students score in these are added to their total LC points. For example, if a student achieves 550 LC points and a score of 200 in the HPAT, their combined LC points is 750 only when the student is applying for courses that require the HPAT. This explains why some courses may have a minimum requirement of 732 LC points, despite 625 being the maximum achievable LC points.

2.4.4 Alternative pathways to college

In Ireland, there are multiple alternative pathways to college, and in this subsection there will be an explanation of the different access routes. Disability Access Route to Education

courses/medicine/

[10] As of 2026, see: https://www.rcsi.com/dublin/undergraduate/medicine/entry-requirements

[11] Langauges other than English and Irish, for example: German, Spanish, Italian, etc.

[12] https://hpat-ireland.acer.org/

[13] https://www.tudublin.ie/study/undergraduate/courses/architecture-tu832/

[14] https://vetting.garda.ie/

[15] https://www.tcd.ie/courses/undergraduate/courses/nursing---general-nursing/

(DARE) [16] is a scheme offering college courses at reduced LC points for students with disabilities. Higher Education Access Route (HEAR) [17] is a scheme offering college courses at reduced LC points and financial aid for students from underrepresented economic or social backgrounds. Students starting college over the age of 23 are categorised as Mature students [18], and their entry into college is not based on LC points, but instead a mature student's entry is based on their educational background, work history and other factors.

2.5 College course fees

In this subsection, there will be an explanation of undergraduate college course fees in Ireland. In Ireland, the majority of undergraduate college courses are publicly funded through the Irish government. As of 2026, the majority of undergraduate college course fees for Irish students cost €2,500 annually [19]. In this context, it is worth introducing the Student Universal Support Ireland (SUSI) [20], Ireland's national awarding authority for college grants. If the combined household income of a student lies under different thresholds [21], the student may qualify for SUSI grants that cover their college course fees and living expenses, giving college opportunities to students from less affluent backgrounds in Ireland.

2.6 Guidance Counsellors

The role of a Guidance Counsellor (GC) has already been described in Section 1.1, however, this section provides additional background information regarding the role of a GC in Irish secondary schools. As legally mandated in Irish Statute Book (1998), every Irish secondary school student has access to a GC to assist them in their educational and career choices. That being said, all GC's do not spend all of their time on guidance counselling. According to Institute of Guidance Counsellors (2024a), 43.6% of GC's are exclusively involved in guidance counselling as part of their role in the school, whereas the other 56.4% combine guidance counselling with other duties, such as subject teaching and school administrative roles.

[16] https://accesscollege.ie/dare/what-is-dare/

[17] https://accesscollege.ie/hear/what-is-hear/

[18] https://www.citizensinformation.ie/en/education/third-level-education/applying-to-college/third-level-courses-for-mature-students/

[19] For instance, see TCD's undergraduate fees: https://www.tcd.ie/courses/undergraduate/fees/

[20] https://www.susi.ie/

[21] for specific circumstances and thresholds, see https://www.susi.ie/eligibility-criteria/income/full-time-undergraduate-income-thresholds-and-grant-award-rates/

2.7 Terminology differences with USA

In Ireland, a "college course" defines the collection of college classes taken over the duration of the student's degree. An example of a college course in Ireland is Computer Science at Trinity College Dublin, encompassing 48 classes taken over the four year degree [22]. In the United States of America (USA), a "college course" refers to a specific class at college, whereas a "college major" is the equivalent terminology for what is defined as a "college course" in Ireland. For example, following the terminology in the USA, you would take a Computer Programming course as part of your Computer Science major. For the purposes of clarity, in this dissertation, the term "college course" will refer to the Irish definition of a "college course". As a result, the terms "college course" and "college major" are seen as equivalents and will be used interchangeably throughout this dissertation.

2.8 Common RS algorithms

TODO - only talk about RS algorithms that are relevant!

2.9 Summary

In summary, this chapter provided the reader with knowledge surrounding the workings of the Irish secondary school and college admissions system. In particular, the LC and LC points system was explained, including the grading system and the way in which subjects are examined at different levels - for example, Higher Level or Ordinary Level. The reader was introduced to the workings of NFQ Levels and the CAO system, developing an understanding into the Irish college admission system. In addition to this, the varying entry requirements into specific courses or colleges were explained. Moreover, the role of GC's were explained in greater detail and this was followed up with a discussion of undergraduate college course fees in Ireland. With this chapter complete, we are better equipped with an understanding behind the Irish secondary school and college admission system.

[22] https://www.tcd.ie/courses/undergraduate/courses/computer-science/

Chapter 3

Literature Review

In this chapter, there will be a review of the literature regarding the fundamentals of college course choice. Ultimately, this section seeks to provide insights to the question: *"What college course should people study?"* As we will discover, the question is incredibly multi-faceted with no straightforward answers. After a thorough review of the fundamentals of college course choice, there will be an examination into closely-related work, including both academic works and real-world products.

3.1 Fundamentals of college course choice

This section will explore the fundamentals of college course choice, and this exploration will follow a narrative as follows. Firstly, we will explore *why* students choose their college and college course by examining the common justifications in their decision. Afterwards, we will examine the external influences, or the *how*, of their college and college course choice, including parent's educational background, gender and others. This equips us with an insight as to *how* and *why* students choose their college course, and the next logical question to ask is: *"So, what college course should people study?"* We will turn our attention to academic literature and popular sentiment in answering this question. Finally, after understanding the *how*, the *why* and the *should* of college course choice, we will examine the features of job and career satisfaction and reverse-engineer these insights to help us answer the all-elusive question: *"What college course should people study?"*

3.1.1 College course justifications

As established in Chapter 1, choosing a college course is a major life decision. As a result, we can assume that students choose a particular college course for a set of reasons. The reasons, or justifications, that students provide for choosing their college course choice will be explored in this subsection.

According to Matthews et al. (2024); Gillis and Ryberg (2021); Mullen (2011), the most common and prominent justifications students cited when choosing their college course can be broadly categorised as: *interest* in the subject area, a sense of *meaning*, *satisfying parent's desires*, *work/life balance*, and *salary expectations* and *job prospects* after graduation. This is by no means an exhaustive list, however, it does cover the most prominent justifications provided by students when justifying their college course choice.

These justifications were all commonly cited across genders and race/ethnicity, however, there were different priorities across gender and race/ethnicity. For instance, Gillis and Ryberg (2021) found that on average, men are more likely to be career-oriented when choosing their college course compared to women, whereas in Matthews et al. (2024), men were found to be more likely to cite *salary expectations* and *job prospects* as their justifications for choosing their college course. Moreover, Goyette and Mullen (2006) found that Asians were more likely to prioritise *salary expectations* in their college course choice, compared to other races/ethnicities. The above examples showcase the different priorities of justifications for college course choice across race/ethnicity and gender. Despite this difference, the papers still conclude that all the above justifications are relevant across the differing genders and races/ethnicities. In other words, despite the findings that, for example, Asians were more likely to prioritise *salary expectations* after graduation, does not imply Asians do not also value other motivations in choosing their college course, such as *interest or passion* in the subject.

3.1.2 College justifications

Choosing a college course is one decision, but a separate and closely related decision is choosing the college itself. For example, a student may settle on studying Computer Science, but the next question to ask is: *Where do I study Computer Science?* In this subsection, there will be an examination into a student's justifications in choosing their college.

Aydin (2015); College MatchPoint (2018); Baharun (2010) have found that students provide a variety of justifications for their college choice, the most prominent justifications include: *prestige* or *reputation* of the college, *location* or *practicality*, *friends* attending the college, *departments* or *academic staff*, *family members* being college alumni, and particular *sports clubs* or *societies*. Once again, this is by no means an exhaustive list, however, it does cover the most prominent justifications of students' college choice.

That being said, there seem to be cultural differences in priorities of college choice justifications. For instance, in Baharun (2010), a study of Malaysian college students, the college's *reputation* was the most important justification in college choice, whereas in Telli Yamamoto (2006), a study of Turkish college students, *departments* and *academic staff* were the most important justification in college choice. While we can clearly observe cultural differences in the most important justifications for college choice, a similar point stands: just because Malaysian students, for example, value college *reputation* as the most important justification in college choice, does not imply Malaysian students do not value other aspects to the college when justifying their college decision, as can be found in Baharun (2010).

3.1.3 External Influences

In subsections 3.1.1 and 3.1.2, we examined the justifications students provided in relation to their college and college course choice. However, in this subsection we will discover that a student does not make this decision in a vacuum. In fact, a student's college and college course choice is shown to be influenced by a variety of external influences. For instance, we will examine the influence of a parent's educational and financial background on a student's college and college course choice. In addition to this, we will explore the influence of gender, race and ethnicity on college course choice. All of the aforementioned influences are being classed under the umbrella term "External Influences" for the sake of clarity.

Parent's educational and financial background

In this discussion, we will explore the influence of a parent's educational and financial background on a student's college and college course choice. According to Leighton and Speer (2023), there is a strong correlation between a student's college course choice and their parent's educational background. According to their findings, students with more educated parents will lean towards college courses with lower early-career earnings, but with faster growth towards higher earnings later in their career. This is likely because students with stronger financial safety nets can afford to pursue college courses with lower early-career earnings in the short-term, and await higher earnings later on in their career.

On the other hand, according to Leighton and Speer (2023), students with less educated parents will lean towards college courses with more defined career paths and higher early-career earnings. This is likely because students with weaker financial safety nets can not afford to pursue college courses with lower early-career earnings in the same way students with stronger financial safety nets can. As a result, students with less educated parents prioritise "safer" college courses with more defined career paths that yield higher early-career earnings. There are analogous findings in Mullen (2011): working-class parents generally encourage their children to attend college courses that directly prepare them for careers, whereas middle class parents generally encourage college courses where their children can explore their interests. Moreover, Mullen et al. (2003) has shown that the parental educational background of a student has a strong correlation with the student's attendance in a postgraduate degree program. The findings above all give credence to the idea that an individual's college course choice is strongly influenced by both their parent's educational and financial backgrounds.

Interestingly, according to Leighton and Speer (2023), they failed to find a strong correlation between the parental income of a student and their college course choice. However, they did find a strong correlation between the parental income of a student and the college they study at. That being said, it is worth mentioning this study was conducted on data obtained from the United States Census Bureau 2009-2018 survey [1]. According to College Board (2025), college undergraduate fees in the USA can range from $4,150 - $45,000 annually, resulting in a much larger financial barrier to entry for American students from less affluent backgrounds. This would explain the strong correlation between parental income and the student's college choice in the USA.

When it comes to Ireland, as discussed in section 2.5, the majority of Irish undergraduate college courses are priced at €2,500 annually, a much more affordable price for undergraduate education compared to the USA. This price not only lowers the financial barrier of entry for Irish students from less affluent backgrounds, but there are also higher education grants [2] for Irish students with income underneath a certain threshold, allowing Irish students from less affluent backgrounds to attend college. Due to the lower financial barrier of entry for Irish undergraduate students in comparison to undergraduate students in the USA, the author would argue that the finding, specifically in Leighton and Speer (2023) that parental income is correlated with a student's college choice, would not be applied so strongly to the Irish context given the lower financial barrier to entry in Ireland.

[1] `https://www.census.gov/programs-surveys/acs`

[2] `https://www.susi.ie/`

That being said, it would be ignorant to assume that parental income is not an influence on college choice for Irish students. For example, according to the QS World University Rankings 2026 [3], the two top-ranked universities in Ireland are in Dublin. According to O'Connor (2025), it is estimated to cost a student an additional €4,000 - €11,000 every year to live away from home in Dublin, compared to someone who lives in Dublin at a family home, clearly demonstrating a limitation imposed on Irish students who cannot afford that kind of money. As a result, we have clearly demonstrated that an Irish student's financial background limits their college choice, which naturally has an influence on their college choice.

Gender

In this discussion we will explore the influence of gender on college course choice. Figure 3.1 visualises the proportional difference of gender across different fields of study in Irish colleges. As is clearly observable, many fields of study are not equal in gender. To give some extreme examples of college course participation in Ireland, according to Higher Education Authority (2024), Fisheries has 100% male participation, and Pre-school education has 97% female participation, demonstrating the large gaps in gender proportions across different fields of study in Ireland.

The above Figure 3.1 and aforementioned statistics clearly demonstrate gender differences across different fields of study, and the next question to ask is: "what are the negative impacts, if any, of studying a college course that features a strong proportion of one gender?" Slattery et al. (2023) conducted a qualitative study on 21 female students studying either Mathematics (66% male), Physics (68% male), Computer Science (74% male) or Engineering (71% male) in Ireland - parenthesised data is from Higher Education Authority (2024). Slattery et al. (2023) found that during their studies, females felt under-valued by their male peers, which negatively impacted their self-beliefs and interest in the course subject matter. In addition to this, they found that females suffered more pressure to perform, and were less likely to engage in the learning environment, showcasing an overall negative experience for females in male-dominated fields of study.

Experiences of isolation and lack of belonging due to gender differences, such as the example given above, are well-known. To provide an example, in I Wish (2025), a survey of Irish secondary school girl's attitudes towards Science, Technology, Engineering and Mathematics (STEM), 65% of girls reported that "Poor gender equality in STEM careers" is a barrier to continuing their studies in STEM. This is quite striking considering

[3] `https://www.topuniversities.com/world-university-rankings?countries=ie`

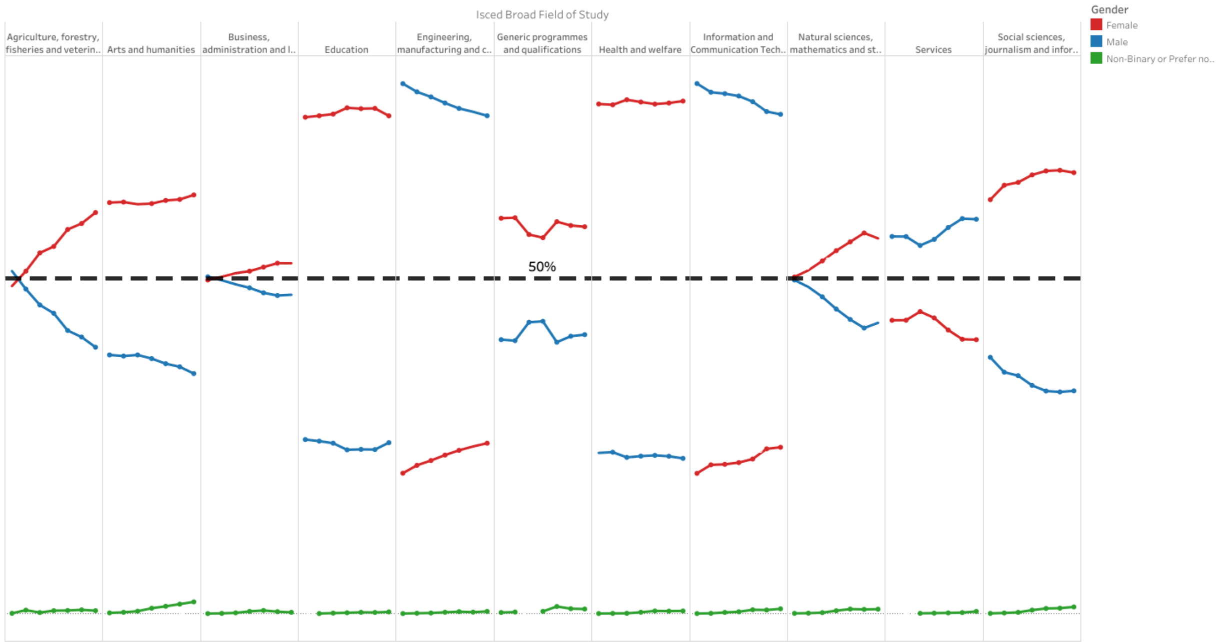

Figure 3.1: Mean gender differences across different fields of study in Irish colleges. Red=Male, Blue=Female, Green=Non-binary or Prefer not to say. Source: Higher Education Authority (2024).

that girls express these viewpoints before they even begin studying in these STEM-related fields. These feelings act as deterrents for girls considering studying in STEM, influencing their college course choice solely on their gender.

A specific example of poor gender equality in STEM was provided in this discussion, however it demonstrates the overall point that gender has a strong influence on college course choice of students in Ireland.

Race and Ethnicity

In this discussion, we will explore the influence of race and ethnicity on college course choice. The author was unable to find statistics for racial or ethnic differences by college course study in Ireland, so statistics from Hinrichs (2015) are used, which was collected from the National Center for Education Statistics in the US Department of Education.

According to analysis from Hinrichs, there are a handful of striking racial or ethnic differences by college course choice. For example, as shown in Figure 3.2, 31% of Asian students study a STEM-related college course, a much higher proportion compared to 11% black, 13% hispanic and 15% white students. An additional example of racial or ethnic differences by college course choice can be found in Figure 3.3, which examines the proportion of race and ethnicity in Public Administration and Social Service: nearly 4% of black students study in this field, a relatively large difference compared to under 1% for Asian, 2.5% hispanic and 1.5% white students studying in this field.

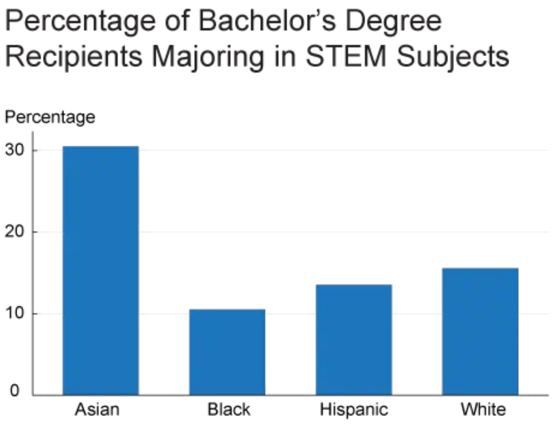

Figure 3.2: Source: Hinrichs (2015).

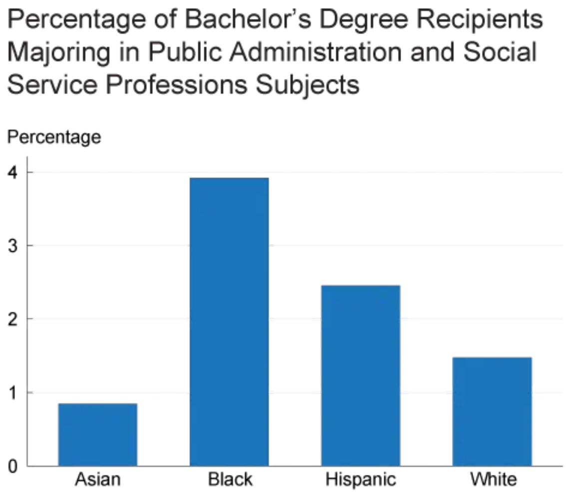

Figure 3.3: Source: Hinrichs (2015).

Further to this analysis, we can examine the work conducted by Dickson (2010), a study performed on administrative data from three Texas Universities (n=127,330). Similar proportions of findings were reported, for example Dickson found that 37% of Science-related college course students were Asian, with 26% white, 24% black and 23% hispanic students being the other prominent racial/ethnic groups. As to why this may be the case, Xie and Goyette (2003) argues that Asian-Americans prioritise "objective" credentials and "hard" skills, which are more difficult to discriminate against in job interviews that are more focused on technical ability, as opposed to job interviews focusing more on "soft" skills. This poses a potential explanation for the relatively hgh proportion of Asians in Science-related college courses compared to other racial/ethnic groups.

Overall, the findings of Hinrichs (2015) and Dickson (2010) demonstrate the differing proportions of race or ethnicity across college course choices in the USA. In this discussion, there was particular focus on the high proportion of Asians in Science-related college courses, for which Xie and Goyette (2003) provided a potential explanation. More generally, this demonstrates the undeniable influence of race and ethnicity on an student's college course choice.

Summary of External Influences

In this subsection, there was an exploration into the external influences of a student's college and college course choice. A student's college course choice is clearly influenced by their personal preferences towards the college and college course choice, as outlined in subsections 3.1.1 and 3.1.2 respectively. That being said, to assume a student's college choice is influenced solely on their personal preferences would be ignorant when one examines the data. In this subsection, there was a discussion into the influence of parent's educational and financial background on a student's college and college course choice. As well as this, there was a discussion on the undeniable influence of gender, race and ethnicity on a student's college course choice. This also involved an exploration into the lack of belonging experienced by students in college courses dominated by another gender. These experiences act as deterrents for students considering these college courses, resulting in an influence on their college course choice, as we saw in the example of low female representation in STEM-related college courses. More generally, this demonstrates the fact that there are indeed external influences on a student's college and college course choice.

3.1.4 Theories of career & college course choice

In the previous subsections, we examined the justifications students provided for picking their college courses, their justifications for picking their college, and the external influences on their college course choice. Ultimately, these subsections attempted to provide insights into the question: *Why do people pick their college courses?* In this subsection, we will turn our attention to academic literature and popular sentiment, and attempt to answer the question: *What college course should people study?*

Holland's theory of career choice

Holland's theory of career choice, first presented in Holland (1959), posits that individual's experience satisfaction when their personality traits match the traits of the role in their work. When talking about personality traits, we are not referring a personality model such as the "Big Five" as proposed in Goldberg (2013). Rather, Holland proposes his own six categories of personality traits present in roles of work. The personality traits are acronymised as RIASEC, and they are defined as:

1. **Realistic:** The doers. They typically enjoy work that involves using their hands, which include making or tinkering with physical objects. Examples of Realistic jobs include Construction Workers and Bakers.
2. **Investigative:** The thinkers. They are intellectually curious and enjoy work that involves observation, research or analysis. Examples of "Investigative" jobs include

Scientists and Philosophers.

3. **Artistic:** The creators. They enjoy expressing themselves creatively and appreciate a lack of structure in solving problems. Examples of "Creative" jobs include Graphic Designers and Actors.

4. **Social:** The helpers. They typically enjoy working with and helping people, and have strong interpersonal skills. Examples and "Social" jobs include Teachers and Nurses.

5. **Enterprising:** The persuaders. They typically enjoy managing people or influencing others, and are often confident and assertive. Examples of "Enterprising" jobs include Lawyers and Managers.

6. **Conventional:** The organisers. They typically enjoy work that involves order and organisation, with strong attention to detail and adherence to rules. Examples of "Conventional" jobs include Accountants and Librarians.

These personality traits in work are alternatively defined as the RIASEC model, an acronym of Holland's six proposed personality traits in work. A visual graphic of Holland's RIASEC model can be seen in figure 3.4. Holland remarks that individuals may exhibit many of the above personality traits simultaneously, and do not necessarily exhibit just one trait. As well as this, the traits are not simply binary and have gradations of strength - in other words, a person may demonstrate strong Investigative qualities, moderate Social qualities and weak Realistic qualities. Holland developed an identification system called "Holland Codes" that involves the first letter of each of his six traits. Holland Codes are essentially used to classify people or jobs under Holland's proposed categories. To give some examples, a Digital Marketer would generally be classed as an "AE" Holland Code, meaning the role exhibits Artistic and Enterprising qualities, as the role involves designing campaigns to influence public perception (Enterprising) and creating visually-appealing displays (Artistic). A Surgeon would generally be classed as an "RIS" Holland Code, meaning the role exhibits Realistic, Investigative and Social qualities, as the role involves strong dexterity with hands (Realistic), deep understanding of human biology (Investigative) and strong patient communication skills (Social). Generally, a Holland Code is no longer than 3 letters.

It is worth mentioning that Holland's proposed personality traits are nuanced and not simplistic. For instance, a biologist may demonstrate "Investigative" qualities through their fascination with human biology, whereas a sociologist may demonstrate "Investigative" qualities through their interest in subcultures. This does not imply that a biologist would be interested in subcultures, or that a sociologist would be fascinated with human biology. In other words, Holland's personality traits manifest themselves differently in different people.

The impact of Holland's theory on the field of Vocational Psychology cannot be overstated - according to Spokane and Cruza-Guet (2005), Holland's theory is the most extensively researched career theory. That being said, some aspects of Holland's theory have been challenged. For instance in Einarsdóttir et al. (2020), they argue the personality model may not appropriately fit the vocational interest structure of Icelanders. In spite of these imperfections, large bodies of research have mostly supported Holland's personality model among high-school students, college students and working adults, according to Nauta (2010), demonstrating the theory's robustness over time.

Holland's theory has become ubiquitous in Guidance Counselling in Ireland. The Career Interests Inventory (CII) [4], a quiz that provides an individual with their Holland Code, is listed by the Irish Department of Education as an acceptable career guidance test in Guidance Counselling practice, according to Department of Education (2024b). In addition to this, the leading career guidance website in Ireland is CareersPortal [5], with under 20 million average active recipients per month [6]. CareersPortal helps users find jobs using an Interest Profiler quiz [7] which is largely inspired by Holland's six personality categories, demonstrating Holland's theory's extensive impact across career guidance and Guidance Counselling in Ireland.

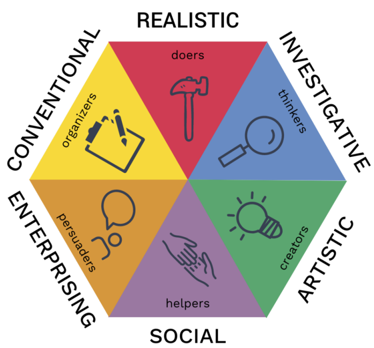

Figure 3.4: Holland's RIASEC model. Source: [8]

[4] https://teamfocus.co.uk/psychometrics/motivation/career-interests-inventory
[5] https://careersportal.ie/
[6] According to https://careersportal.ie/about-us
[7] https://careersportal.ie/members/interests_check.php

Life-Span, Life-Space Theory

Despite the evident success and utility of career fit theories' such as Holland's, there are limitations to these kinds of theories. Most notably, matching an individual's traits to a job's traits involves the individual making a decision at a particular point in time. Super (1980) argues that an individual's career is not one static decision. Rather, Super poses that career development is a lifelong process with a series of developmental stages, in which the individual's self-concept evolves over their lifespan. In this view, an individual's career is the cumulative outcome of many decisions taken at different points in time. Even in the case where an individual's studied college course is not relevant to their job - in which 23% of Irish graduates reported in Higher Education Authority (2018) - under Super's theory, that decision is still encompassed as part of the individual's career.

Super coined this as Life-Span, Life-Space Theory (LSLST). The "Life-Span" refers to the lifelong developmental process of career, whereas "Life-Space" refers to the jobs and roles that occupy the individual at different stages in their career, according to the definition from Hartung (2013). In spite of the prominence of LSLST, it has faced criticism for failing to account for the dynamic careers of women, which are often interrupted by caregiving, according to Fitzgerald and Betz (1994). It is worth noting that Holland's theory is not at odds with Super's theory, rather Super intended to complement the established utility of Holland's theory, according to Greenhaus and Callanan (2006).

The Passion Hypothesis

One of the most popular sentiments when it comes to career advice is to *"Follow your Passion"*. This sentiment was birthed in Bolles (1972), a self-help book for job-hunters, which sold over ten million copies [9]. The sentiment gained a surge in popularity when Steve Jobs, the co-founder of Apple [10], famously claimed in his commencement speech at the 2005 Stanford University Graduation: *"The only way to do great work is to love what you do"* [11]. Jobs' speech garnered a large amount of popularity, amassing over 48 million views [12] on YouTube, demonstrating the sentiment's widespread appreciation.

Newport (2016) coins this as the Passion Hypothesis - the idea that to find great work, you simply find out what you love and follow those interests. As appealing as "Fol-

[9] As of 2016 according to: `https://shorturl.at/jiHn2`
[10] `https://www.apple.com/`
[11] `https://www.youtube.com/watch?v=UF8uR6Z6KLc`
[12] As of March 2026.

low your Passion" sounds, the advice has many limitations. The underlying implication of this advice is that passions are pre-existing, and neglects the idea that passions can be developed. This then prompts the question: *"Can passion be found or developed at work?"*, which is answered in O'Keefe et al. (2018). Following the definition in Dweck (2006), In O'Keefe et al.'s, experiment, the participants were broadly categorised into "fixed theorists", who believed that personal interests are relatively fixed, and "growth theorists" who believed personal interests are developed. According to O'Keefe et al.'s findings, the fixed theorists were more likely to dampen their interests in areas outside of their existing interests, suggesting a level of narrow-mindedness for the fixed theorists. As well as this, when engaging in a new interest became difficult, the interest dropped significantly more for fixed theorists in comparison to growth theorists, suggesting that growth theorists have more endurance when encountering inevitable difficulties when pursuing their interests.

Endorsing the Passion Hypothesis implies that one's passions are pre-existing, and fails to recognise that passions can be nurtured at work, as is similarly found in Chen et al. (2015). While the sentiment is not a theory according to scientific definition [13]- which is why it was coined as the Passion Hypothesis, as opposed to the Passion Theory - it is still worth discussing in this subsection. In spite of the sentiment's appeal and popularity, we examined the sentiment's limitations as career advice.

Summary of Theories of career & college course choice

In this section, we turned our attention to academic literature and popular sentiment in answering the question: *What college course should people study?* An exploration of Holland's theory of career choice was provided, a ubiquitous and transformative career choice theory. In spite of Holland's theory's established robustness in the literature, it fails to account for the dynamic, lifelong nature of careers. The work of Super nicely fits in, as Super conceptualises career in this manner under Life-Span Life-Space theory (LSLST). This theory emphasises the notion of career as a developmental process in which an individual evolves in their self-concept. Super developed LSLST to complement the growing influence of Holland's theory. We can then turn our attention away from academic literature to popular sentiment - in which an exploration into the appealing advice "Follow your passion" was given. On the whole, this section provided the reader with a solid understanding into the differing theories of career and college course choice.

[13]Following the definition according to `https://www.fieldmuseum.org/blog/what-do-we-mean-theory-science`

3.1.5 Job & career satisfaction

In subsection 3.1.4, we turned our attention to academic literature and popular sentiment, to provide insights to the question: *What college course should people study?* In this subsection, rather than finding out what college course people should study by looking forward, we will take a look back and examine the satisfaction of individuals in their jobs and careers. The insights gleamed from examining job and career satisfaction will help us answer the above question. It is worth mentioning that jobs are distinct to careers; following the definition from Super (1980), career is the cumulative outcome of an individual's vocational decisions over their lifespan. On the other hand, a job is at a specific stage in an individual's career. While jobs are distinct to careers, we will discover that the traits of job and career satisfaction are very similar, so in this discussion, the traits of job and career satisfaction will be explored together.

Before we begin our examination on the factors of job & career satisfaction, it is worth taking some time to examine what individuals value most from their work, and a relevant paper for this discussion would be Harpaz (1990). In spite of the paper's age, the paper has particular relevance as it was conducted across 7 countries and 3 continents with n=8192. According to Harpaz' findings, the two most prominent work values were *Interesting Work* and *Salary*. These values were consistent across gender, age and nationality. As well as this, the next most prominent work values were *Colleague Relationships*, *Job Security* and *Autonomy*. With an overview of what individuals most value in their jobs, we shall now examine the literature to investigate whether these values contribute substantially towards job satisfaction. It is worth noting that some, not all, of these work values will be examined in detail for the sake of brevity.

Before we examine these work values in detail, essential to the discussion of job satisfaction is Herzberg's Two-factor theory as outlined in Herzberg (1959). Herzberg's theory posits that job satisfaction is influenced by two factors, which are coined as *Motivators* and *Hygiene Factors*. Motivators *lead* to job satisfaction, whereas Hygiene Factors *prevent* job dissatisfaction. Examples of Motivators include *Interesting Work*, *Achievement*, *Autonomy* and a sense of *Meaning*, and examples of Hygiene Factors include *Salary*, *Job Security*, *Colleague Relationships* and *Status*. A graphic of Herzberg's two-factor theory can be seen in figure 3.5. Essentially, Herzberg's theory claims that job satisfaction has different factors to job dissatisfaction. For example, an individual may be in a high-paying, respectable job with positive colleague relationships, however the individual may not derive a sense of meaning or have interest in their job. According to Herzberg's theory, the individual would not be *dissatisfied* with their job - after all, the job does tick some boxes - although the individual would not be incredibly *satisfied* with their job given the absence of Motivators.

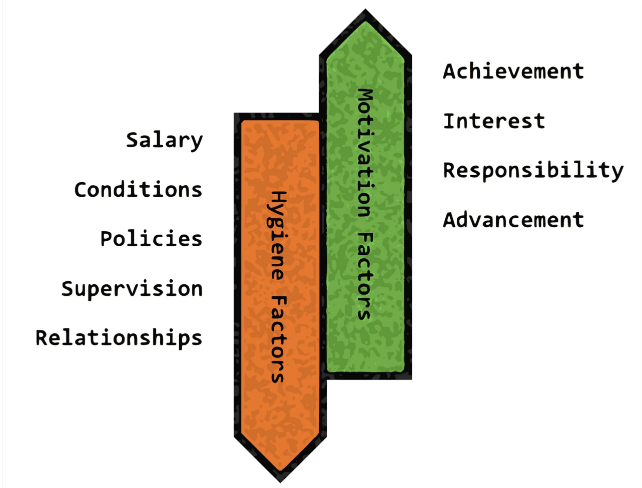

Figure 3.5: Herzberg's Two-factory theory. Source: [14]

Many of Herzberg's claims are supported in the literature. For example, in Hoff et al. (2018), a meta-analysis across 105 studies, they found statistically significant positive correlation between interest-job fit and job satisfaction. Similary, Nye et al. (2012) found that interest-job fit - or more precisely, Holland's vocational personality traits - were positive indicators for job performance and persistence in work. As mentioned previously, *Interesting Work* was one of the two most prominent work values of individuals according to Harpaz (1990), suggesting that individuals strongly value a factor that significantly contributes towards their job satisfaction. On the whole, these findings support the idea that having alignment between the individual's interests and and their job's work is vital for job satisfaction.

In spite of *Salary* being one of the two most prominent work values according to Harpaz (1990), Herzberg claims that salary is a Hygiene Factor - in other words, sufficient salary in a job prevents job dissatisfaction, but it does not necessarily lead to job satisfaction. According to a meta-analysis conducted by Judge et al. (2010), they found a weak relationship between salary and job satisfaction, giving credence to Herzberg's theory with regards to salary being a Hygiene Factor (prevents job dissatisfaction) as opposed to a Motivator (provides job satisfaction). This is a particularly striking finding given that salary was one of the two most prominent work values according to Harpaz (1990).

We can further this discussion of salary in work by making reference to Ward (2024), in which they discuss a common conundrum between a meaningful job or a job with a high salary. According to their findings, individuals will consistently choose a job with a high salary and a low sense of meaning, compared to a job with a low salary and high sense of meaning. In spite of the findings of a weak relationship between salary and job satisfaction, the author would argue that prioritising salary over a sense of meaning in a job, or other work values, is understandable. According to National Youth Council of Ireland (2025), 3 of 5 Irish adults under 25 years old are considering emigrating due to the rising cost of living in Ireland, clearly demonstrating the importance of salary in jobs.

In summary, this section introduced Herzberg's two factor theory which posits that job satisfaction can be separated into Motivators, which lead to job satisfaction, and Hygiene Factors which prevent job dissatisfaction. We examined the most prominent work values of individuals, with a particular focus on Salary and Interesting work. The literature finds a significant relationship between job satisfaction and interest, althought on the other hand, the literature finds a weak relationship between job satisfaction and salary. That being said, it was argued that prioritisation of salary in jobs is understandable given the rising cost of living in Ireland.

3.1.6 Summary and synthesis of findings

This subsection will provide a brief summary of this section's key findings, and then present a synthesis of the findings.

In summary, we first examined the common justifications students provided for their college course choice, which included *interest*, *salary* and a sense of *meaning*. Secondly, we examined the common justifications students provided for their college choice, which included *prestige/status* and *location/practicality*. As well as this, students' college course choices were found to have undeniable external influences, the most notable of which are their parent's educational and financial background, gender and race/ethnicity. Understanding the background to a student's justifications and external influences are key to understanding how and why students choose their college course. As part of our exploration in the fundamentals of college course choice, there was an examination into differing theories of career and college course choice. This included a description of Holland's 6 personality traits in work, also known as the RIASEC model, a transformative theory that is ubiquitous in Irish Guidance Counselling. In addition to this, we examined the literature of job satisfaction and discussed the strong relationship between interest-fit and job satisfaction, and the weak relationship between salary and job satisfaction.

As described in this section's introduction, this section attempted to provide insights to the question: *"What college course should people study?"* As is evident from the coverage of this literature, the answer to this question, if there is one, is not straightforward. Despite this, this dissertation will attempt an attempt to answer this question by synthesising this section's findings.

So, according to this dissertation's opinions and findings, *what college course should people study?* Given the significant relationship between interest-job fit and job satisfaction, and the fact that many individuals value interesting work in their college course jobs, an individual should consider studying a college course that they either **find interesting,** or can see themselves **developing an interest in.** As well as this, given the robustness and ubiquity of Holland's theory of career choice, or the RIASEC model, an individual should consider studying a college course that aligns with their **RIASEC model.** Moreover, given the aforementioned rising cost of living in Ireland, students may also consider college courses for their **salary** or **job prospects.** Many students expressed interest in college courses or careers that are meaningful, so a student may also consider a college course for their sense of **meaning** or **impact.** Beyond the college course, a student should also consider the actual **college** at which to study their chosen course, which may impact their decision depending on the availability of the college course at different colleges.

This is of course, a simplified answer to a very multi-faceted question, and the author would like to acknowledge that this answer serves more as a guideline, as opposed to a "one-size-fits-all" answer. In addition to this, the factors provided in this answer were deemed to be the most prominent or important factors given the reviewed literature, however, as is evident from our prior review, there are many reasons as to why a student may be satisfied with their college course choice, and this reason may be beyond the scope of the provided answer. For example, a student may choose to study a particular course primarily because they are seeking positive work/life balance.

The reader may notice that a handful of the previously discussed topics, such as gender, did not feature in this dissertation's attempt at answering the question: *what college course should people study?* The author would argue that even if some factors, for example gender, do not feature in this dissertation's answer attempt, it is still worth discussing the clear influence these factors have on college course choice. Having an awareness of these factors, even if they do not explicitly feature in the final answer attempt, are still useful in understanding the nature of college course choice, and having an implicit, intuitive understanding in answering the all-elusive question: *what college course should people study?*

3.2 Closely-Related Work

As outlined in section 1.1, choosing a suitable college course is a multi-faceted, difficult decision for students. The college course decision is personally confronting for students and there are a variety of external influences on their college course choice, as is evident from the examinations in subsections 3.1.1 and 3.1.3. Technologies, not limited to a Recommender System (RS), have been used to assist students in finding suitable college courses for them. In this section, we will examine the academic literature and real-world products for technologies that have assisted students in their college course decision-making process. We will start with the simplest technologies, and work our way up through more sophisticated solutions to college course choice for students.

3.2.1 Simple filters

CareersPortal [15] and Qualifax [16] are the two leading college course databases in Ireland [17] [6]. The simplest technology for assisting Irish student's in their college course selection is to use the preset filters on these websites. CareersPortal allows users to filter by multiple features shown in Figure 3.6, which includes filtering by career sector (see Figure 3.7), and career interests [18] (see Figure 3.8). After applying some of these filters, a user would be presented with a result as shown in Figure 3.9.

[15]`https://careersportal.ie/`

[16]`https://www.qualifax.ie/`

[17]Qualifax has over 500,000 users annually. source: `https://www.annertech.com/case-studies/qualifax-making-information-easy-irelands-learners`

[18]CareerPortal's career interests are explained in Chapter 6 **TODO**

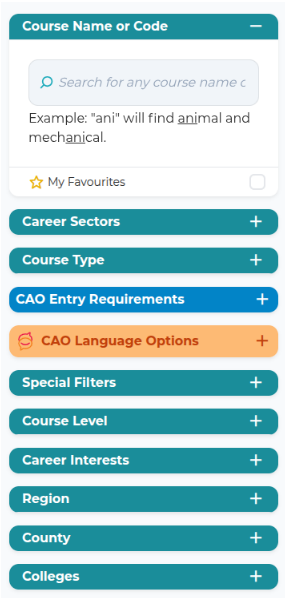

Figure 3.6: CareerPortal's college course filters.

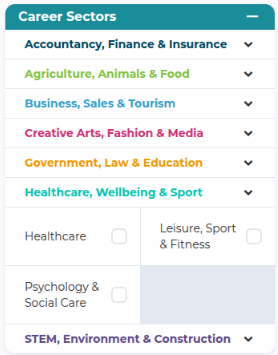

Figure 3.7: CareerPortal's career sector filter.

Figure 3.8: CareerPortal's career interests filter.

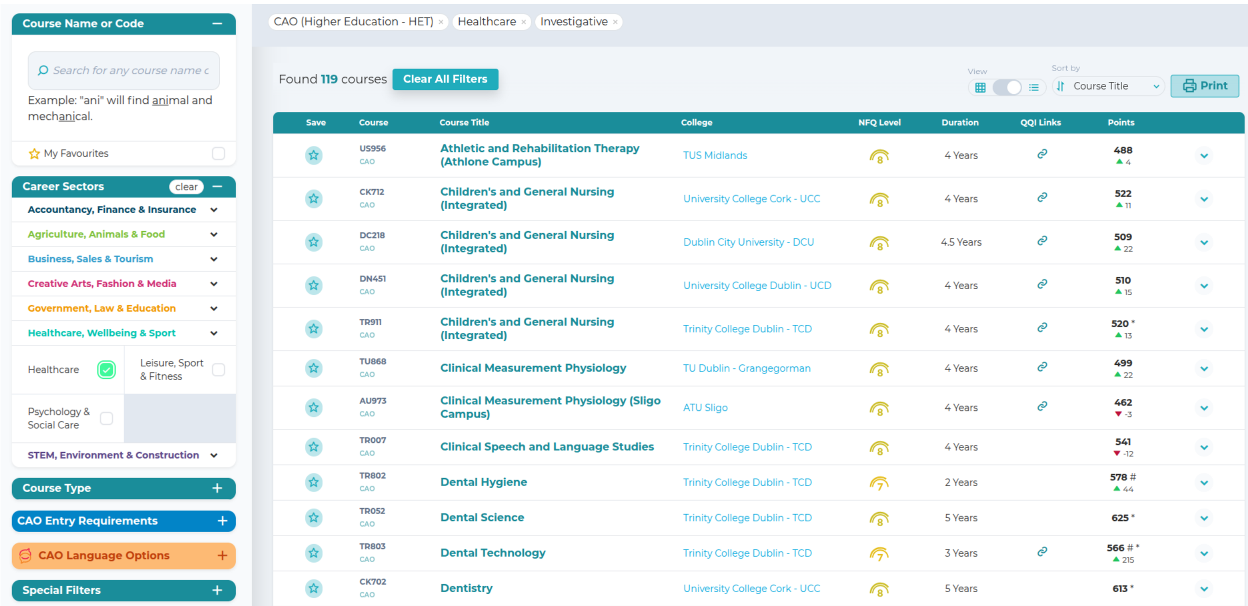

Figure 3.9: A sample result of what users would see after applying some filters to a course search in CareersPortal.

Qualifax operates similarly: they allow user to filter their college course database by numerous filters, including by college course category (Figure 3.10). After applying some of these filters, a user would be presented with a result as shown in Figure 3.11.

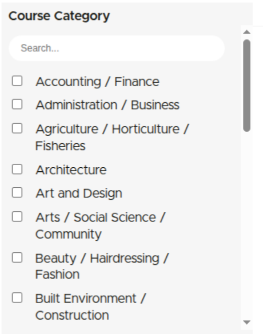

Figure 3.10: Qualifax' college course category filter.

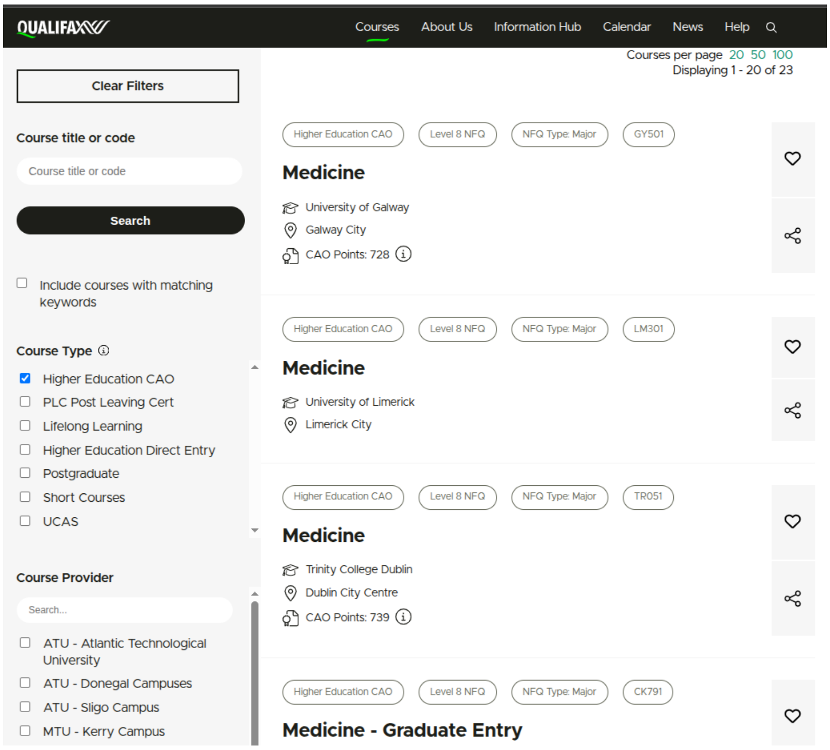

Figure 3.11: A sample result of what users would see after applying some filters to a course search in Qualifax.

Simple filters in college course search, such as the CareersPortal and Qualifax websites, are useful as they allow students to scan a large variety of courses. However, the strength of these filters are also it's greatest weakness - the quantity of choice. Even after applying a handful of filters, for example in Figure 3.9, a user is still presented with 119 courses. This is a large quantity of courses, which may result in a phenomenon known as overchoice, coined in Toffler (1970). According to Toffler, overchoice is a paradoxical phenomenon where choosing between a large variety of options can be detrimental to decision-making processes. As a result, we find that simple filtering technologies present large quantities of college course options, however, this does not necessarily make the decision-making process easier due to the phenomenon of overchoice. Long et al. (2025) demonstrated that smaller sets of suggestions are often equally as effective as larger suggestion sets [19], calling for technologies to apply *personalisation*, pruning their results and presenting more tailored college courses to the student, setting the scene for our next subsection.

[19]For specificity, this research was conducted in the context of Recommender Systems, however the findings are still appropriately applied in this case.

3.2.2 College course quizzes

In this subsection, we will examine technologies that provide more personalisation to a user in the context of college course recommendations. More specifically, there will be a discussion on online quizzes that provide users with a more tailored, personalised list of college course recommendations. As a side note, seeing as these are real-world products, we will not have an insight into the recommendation algorithms behind these quizzes, we will only have access to their front-facing user interface.

For this discussion, we will examine the college major quizzes offered by MeetYourClass [20], SuperProf [21], Anderson University [22] and Marquette University [23]. It is worth noting that these quizzes have an American focus, however their results can be applied more generally.

These quizzes ask the user close-ended questions, and generally the questions are tasked with discovering the student's interests in different areas of study. Examples of the kinds of questions that get asked are seen in Figures 3.12 and 3.13.

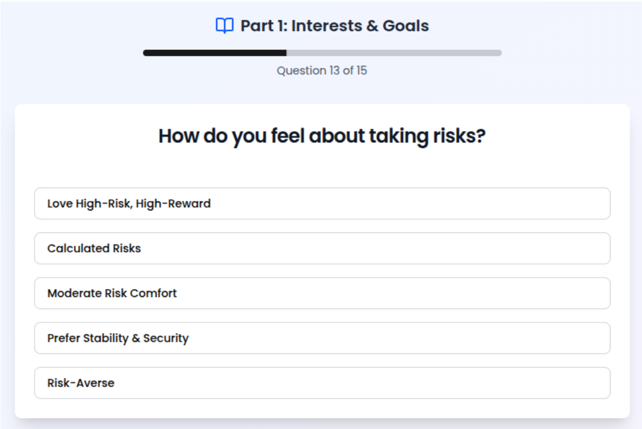

Figure 3.12: A sample question provided on MeetYourClass' college major quiz.

Figure 3.13: A sample question provided on Anderson University's college major quiz.

After answering all the questions, the quizzes provide the user with a suggested list of college majors based off the answers provided to the questions. Examples of results are given in Figure 3.14 and 3.15.

[20] https://www.meetyourclass.com/what-should-i-major-in
[21] https://www.superprof.com/blog/college-major-quiz/
[22] https://anderson.edu/persona-quiz/
[23] https://www.marquette.edu/academics/majors/choose-your-major/

Figure 3.14: A sample results page after finishing the college major quiz on Anderson University.

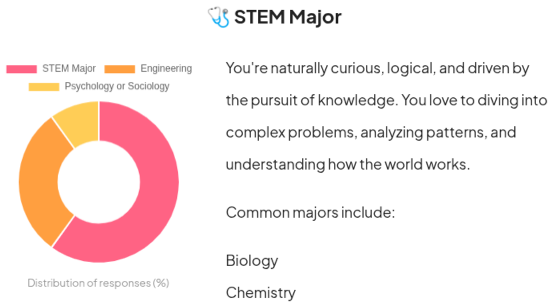

Figure 3.15: A sample results page after finishing the college major quiz on SuperProf.

In spite of the evident utility of these quizzes, they do pose some limitations. Firstly, considering the importance of the college course decision process, these quizzes are generally very short - the four examples provided had 10-25 questions each. If we take Marquette University's quiz as an example: 79 college majors are offered [24], and 10 questions may not be considered sufficient to gauge a student's interest appropriately across the 79 college majors. In other words, the length of the quizzes imply they are trivialising and simplifying the solution to the problem.

In addition to this, generally the results of these quizzes may not refer to specific college majors, although they may refer to broader fields of study. For example as we see in Superprof's results in figure 3.15, STEM is being suggested to the user, however STEM encompasses a wide range of college majors including Mathematics and Engineering - similar, but still different fields of study. While this may guide the user toward narrowing their college major choices, there is still a lack of specificity in the college major recommendations.

However, if one wants to go beyond broader fields of study and have more specificity in their college course recommendations, the technology is required to have knowledge of specific college courses being offered at specific colleges. This introduces a new requirement of the technology having up-to-date, accurate college course information being offered at multiple different colleges. Moreover as mentioned in chapter 2, the college admissions system and entry requirements differ by country, and clearly the Irish college system differs to the American college system that these quizzes refer to. Seeing as this dissertation focuses on Irish college course recommendations, these requirements call for a technology with a working knowledge of the Irish college courses and college system,

[24] According to `https://www.marquette.edu/academics/majors/`

and that technology suggests specific, personalised college courses in Ireland.

The technology described above already does exist: FindMyCollegeCourse (FMCC) [25]. To the best of our knowledge, this is the only online quiz that recommends college courses in Ireland. FMCC addresses many of the aforementioned limitations of other college major quizzes: the quiz has over 100 questions, demonstrating much more breadth in capturing student's interests and preferences. Additionally, FMCC recommends specific Irish college courses as opposed to broad fields of study, and the college course information is up-to-date and accurately maintained - a challenge given the volume of new courses being added each year; see footnote 6. Naturally, the quiz is focused in the Irish college domain, which is also the focus of this dissertation. That being said, there are still some limitations and improvements which can be applied, although a more thorough discussion of this will be provided in section 3.3.

3.2.3 College course Recommender Systems

In Chapter 1, there was a reference to the Recommender System (RS) technology. In this subsection, we will turn our attention from real-world products to the academic literature, and examine the intersection of RS's and college course recommendations in the literature. We will break down our investigation into two fundamental questions:

1. What information is the user being questioned in order to make relevant college course recommendations?

2. What algorithms are behind the RS' to generate college course recommendations?

User information

In order for an RS to provide personalised college course recommendations, the RS must gather some information from the user. The next logical question to ask is: *"What information should we gather from the user?"* However, as was evident from our investigation in section 3.1, we found that answering the question *"What college course should people study"* to be multi-faceted, encompassing many factors. In this discussion, we will examine the types of information that other works in the literature have gathered off the user in order to provide personalised college course recommendations.

[25] `https://findmycollegecourse.ie/`

When we examine Qamhieh et al. (2020); Lahoud et al. (2023); Ghuge et al. (2023); Bandhu et al. (2024), generally we can observe the following information being profiled off students in order to make personalised college course recommendations:

- *academic interests*, e.g. interest in Mathematics, Biology, etc.
- *high-school grades*
- *hobbies or extracurriculars*
- *gender*
- *personality type*, specifically the Myer-Briggs Type Indicator (MBTI) Myers et al. (1962).

TODO - should I discuss the validity of these information profiling? e.g. i think mbti is noise and unnecessary, same for gender and hobbies and high-school grades but maybe this makes more sense in the design chapter because that's where I will justify my design decisions? ask @owen!

RS Algorithms

In this discussion we will examine the various algorithms behind the referenced literature. An understanding of the common RS algorithms, as discussed in section 2.8, will be useful to the reader. However, a basic understanding of common Machine Learning (ML) algorithms is assumed.

In Qamhieh et al. (2020)[26], the Content-based Filtering algorithm was utilised and was powered by a mixture of a weighted average score and an if-then ruleset. Ghuge et al. (2023) utilises a hybrid RS with Collaborative Filtering and Content-based Filtering algorithms being used, powered by the AdaBoost algorithm as introduced in Freund and Schapire (1997), whereas in Lahoud et al. (2023), they employ a wide range of RS algorithms, and they found the best results with Content-based Filtering and Knowledge-Based Filtering. In Bandhu et al. (2024), they employ the Content-based Filtering approach, powered by numerous ML algorithms including Random Forest (Breiman (2001)), Support Vector Machines (Cortes and Vapnik (1995)), Naive Bayes (Bayes (1968)) and K-nearest neighbours (Cover and Hart (1967))- in their case, they found the best results with the Random Forest algorithm.

im not really sure what else to write here or if there are more things i can address - ask owen!

[26] *Note:* this paper recommends engineering fields of study as opposed to college courses, however the paper is arguably still relevant

3.3 Research Gap

3.4 Chapter Summary

Chapter 4

Initial Prototypes

Chapter 5

Design

It is worth noting that there are no intentions behind this RS to replace the vocational support role of a GC. Choosing a college course is a difficult and personally confronting decision, and as mentioned in Institute of Guidance Counsellors (2016), GC's are trained to additionally support a student's social and personal needs, allowing the GC to provide a student with empathetic and human-centered support - an aspect that a traditional RS would fail to effectively convey.

Furthermore, there are no intentions behind this RS to entirely replace the student's college course decision-making process. As we will discover in Chapter 3, the decision of a college course is incredibly nuanced. It is the author's opinion that no system can or should make that decision entirely for an individual. The intention behind this RS is to *complement*, rather than replace, the student's college course decision-making process, assisting the student in exploring a variety of career paths that may be relevant to them.

5.1 Relevant information to question

Interviews with expert guidance counsellors were conducted as part of this research, and a qualitative analysis of these interviews are provided in Section 5.1.1.

5.1.1 Interviews with guidance counsellors

5.2 Problem Formulation

This section should provide the reader with an overall description of the problem that will be addressed in the dissertation. In contrast to a generic discussion of the dissertation topic in the Introduction chapter, this section should provide a detailed discussion of the problem that has been identified based on the existing work that has been discussed in the preceeding chapter.

In some dissertations, it may make sense to convert this section into a short chapter of its own which follows the discussion of the existing work and preceeds the discussion of the work of the dissertation.

5.2.1 Identified Challenges

This section should present a short description of the gaps in the existing work and the relationship of these gaps to the work described in this dissertation.

5.2.2 Proposed Work

This section should provide a thorough description of the problem and an overview of the work proposed to address the problem.

5.3 Overview of the Design

A description of the approach that addresses the problem identified above.

5.4 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 6

Implementation

Guess what? At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

6.1 Overview of the Solution

The code in listing **??** is a demonstration how to include a file with code into the template.

6.2 Component One

The code in listing 6.1 is a demonstration how to include code in the template.

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 6.1: Second Lengthy caption

6.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 7

Methodology

Chapter 8

Evaluation

The Evaluation chapter should present a comparison of the work that forms the basis of the dissertation and existing work. At a higher level, it should demonstrate an awareness of the relationship of the dissertation work to the research area that it is based in.

8.1 Experiments

In the case where experiments have been carried out, the experimental setup and the values that were defined for the variables need to be presented in a table e.g. table 8.1.

Column 1	**Column 2**	**Column 3**
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 8.1: Caption that explains the table to the reader

8.2 Results

Figures that present results such as figure **??** need to display descriptions of the axes, the units and scales of the measurements, statistical values, etc. Where measurements were taken from experiments, error bars or confidence intervals need to be provided to give the reader an indication of the spread of the measurements.

8.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary that presents the outcome of the chapter in a short, accessible form.

Chapter 9

Conclusions & Future Work

This chapter should summarize the work presented in the dissertation and discuss the conclusions that can be drawn from the work and the results presented in chapter **??**.

9.1 Future Work

The section may present a list of items that were beyond the scope of the dissertation.

Bibliography

Aydin, O. (2015). University choice process: A literature review on models and factors affecting the process. *Yuksekogretim Dergisi*, 5.

Baharun, R. (2010). Institutional factors influencing students' college choice decision in malaysia: A conceptual framework. *International Journal of Business and Social Science*.

Bandhu, K. C., Litoriya, R., Khatri, M., Kaul, M., and Soni, P. (2024). A novel approach for better career counselling utilizing machine learning techniques. *Wireless Personal Communications*, 138(4):2523–2560.

Bayes, T. (1968). Naive bayes classifier. *Article Sources and Contributors*, pages 1–9.

Bolles, R. N. (1972). *What color is your parachute?* Ten Speed Press Berkeley, CA.

Breiman, L. (2001). Random forests. *Machine learning*, 45(1):5–32.

Central Applications Office (2025). Cao application statistics - february 2025. `https://www2.cao.ie/app_stats/pdf/appstats01feb2025.pdf`. Accessed: 2026-03-11.

Chen, P., Ellsworth, P. C., and Schwarz, N. (2015). Finding a fit or developing it: Implicit theories about achieving passion for work. *Personality and Social Psychology Bulletin*, 41(10):1411–1424.

College Board (2025). Trends in college pricing 2025. `https://research.collegeboard.org/media/pdf/Trends-College-Pricing-2025.pdf`. Accessed: 2026-03-11.

College MatchPoint (2018). How students decide which college to attend. `https://www.collegematchpoint.com/college-matchpoint-blog/2018/4/making-the-right-choice-how-students-decide-which-college-to-attend`. Accessed: 2026-03-11.

Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine learning*, 20(3):273–297.

Cover, T. and Hart, P. (1967). Nearest neighbor pattern classification. *IEEE transactions on information theory*, 13(1):21–27.

Department of Education (2024a). Department of education press release: Ministers congratulate almost 61,000 students receiving their leaving certificate examination results today. `https://www.gov.ie/en/department-of-education/press-releases/ministers-congratulate-almost-61000-students-receiving-their-leaving-certificate-` Accessed: 2026-03-11.

Department of Education (2024b). List of assessment instruments/tests for guidance or for additional and special educational needs. `https://assets.gov.ie/static/documents/list-of-assessment-instrumentstests-for-guidance-or-for-additional-and-special-e` `pdf`. Accessed: 2026-03-11.

Dickson, L. (2010). Race and gender differences in college major choice. *The Annals of the American Academy of Political and Social Science*, 627(1):108–124.

Dweck, C. S. (2006). *Mindset: The new psychology of success*. Random house.

Einarsdóttir, S., Björnsdóttir, M. D., and Lerkkanen, J. (2020). 12. cross-cultural validation of assessment instruments used in career counselling and guidance in the nordic countries. *Career and career guidance in the Nordic countries*, pages 177–189.

Fitzgerald, L. F. and Betz, N. E. (1994). Career development in cultural context: The role of gender, race, class, and sexual orientation.

Freund, Y. and Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of computer and system sciences*, 55(1):119–139.

Ghuge, M., Kamble, T., Mandrawliya, A., Kumari, A., and Raikwar, V. (2023). Envisioning tomorrow: Ai powered career counseling. In *2023 3rd International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)*, pages 377–383. IEEE.

Gillis, A. and Ryberg, R. (2021). Is choosing a major choosing a career or interesting courses? an investigation into college students' orientations for college majors and their stability. *Journal of Postsecondary Student Success*, 1(2):46–71.

Goldberg, D., Nichols, D., Oki, B. M., and Terry, D. (1992). Using collaborative filtering to weave an information tapestry. *Communications of the ACM*, 35(12):61–70.

Goldberg, L. R. (2013). An alternative "description of personality": The big-five factor structure. In *Personality and personality disorders*, pages 34–47. Routledge.

Goyette, K. A. and Mullen, A. L. (2006). Who studies the arts and sciences? social background and the choice and consequences of undergraduate field of study. *The Journal of Higher Education*, 77(3):497–538.

Greenhaus, J. and Callanan, G. (2006). Super's career development theory. *Encyclopedia of Career Development*, pages 790–792.

Harpaz, I. (1990). The importance of work goals: An international perspective. *Journal of international business studies*, 21(1):75–93.

Hartung, P. J. (2013). The life-span, life-space theory. *Career development and counseling: Putting theory and research to work*, page 85.

Herzberg, F. (1959). *Motivation to work*. Routledge.

Higher Education Authority (2018). Graduate outcomes: Honours degree graduates. `https://hea.ie/statistics/graduate-outcomes-data-and-reports/graduate-outcomes-2018/honours-degree-graduates/`. Accessed: 2026-03-11.

Higher Education Authority (2024). Key facts and figures for ireland's publicly-funded higher education institutions. `https://hea.ie/statistics/data-for-download-and-visualisations/key-facts-figures-report/`. Accessed: 2026-03-11.

Hinrichs, P. L. (2015). Racial and ethnic differences in college major choice. `https://www.clevelandfed.org/publications/economic-trends/2015/et-20150331-racial-and-ethnic-differences-in-college-major-choice`. Accessed: 2026-03-11.

Hoff, K., Wee, C., Song, Q. C., Phan, W. M. J., and Rounds, J. (2018). Meta-analysis of vocational interests and job satisfaction: Challenging a common belief. *Academy of Management Proceedings*, 2018:12220.

Holland, J. L. (1959). A theory of vocational choice. *Journal of counseling psychology*, 6(1):35.

I Wish (2025). I wish 2025 survey report. `https://www.iwish.ie/wp-content/uploads/2025/10/I-Wish-2025-Survey-Report-Final.pdf`. Accessed: 2026-03-11.

Institute of Guidance Counsellors (2016). Guidance counselling core competencies for professional practice. https://igc.ie/wp-content/uploads/2020/04/Guidance-Counselling-Core-Competencies-Professional-Practice-1.pdf. Accessed: 2026-03-11.

Institute of Guidance Counsellors (2022). Igc national survey 2022/2023. https://igc.ie/wp-content/uploads/2024/03/IGC-National-Survey-2022-2023-final.pdf. Accessed: 2026-03-11.

Institute of Guidance Counsellors (2024a). Igc national survey 2024/2025. https://igc.ie/wp-content/uploads/2025/12/IGC-National-Survey-2024-2025-final.pdf. Accessed: 2026-03-11.

Institute of Guidance Counsellors (2024b). National strategic plan 2024-2027. https://igc.ie/igc-nationalstrategic-plan-2024-2027/. Accessed: 2026-03-11.

Irish Statute Book (1998). Education act 1998. https://www.irishstatutebook.ie/eli/1998/act/51/section/9/enacted/en/html#:~:text=c%20ensure%20that%20students%20have%20access%20to%20appropriate%20guidance%20to%20assist%20them%20in%20their%20educational%20and%20career%20choices%2C. Accessed: 2026-03-11.

Judge, T. A., Piccolo, R. F., Podsakoff, N. P., Shaw, J. C., and Rich, B. L. (2010). The relationship between pay and job satisfaction: A meta-analysis of the literature. *Journal of vocational behavior*, 77(2):157–167.

Lahoud, C., Moussa, S., Obeid, C., Khoury, H. E., and Champin, P.-A. (2023). A comparative analysis of different recommender systems for university major and career domain guidance. *Education and information technologies*, 28(7):8733–8759.

Leighton, M. and Speer, J. (2023). Rich grad, poor grad: family background and college major choice. *Institute of Labor Economics (IZA)*.

Long, X., Sun, J., Dai, H., Zhang, D., Zhang, J., Chen, Y., Hu, H., and Zhao, B. (2025). The choice overload effect in online recommender systems. *Manufacturing & Service Operations Management*, 27(1):249–268.

Matthews, K. N., Edmondson, D. R., and Matthews, L. (2024). Why students select their college major: An investigative study. *Atlantic Marketing Journal*, 13(1):11.

Mullen, A. L. (2011). *Degrees of inequality: Culture, class, and gender in American higher education*. JHU Press.

Mullen, A. L., Goyette, K. A., and Soares, J. A. (2003). Who goes to graduate school? social and academic correlates of educational continuation after college. *Sociology of education*, pages 143–169.

Myers, I. B. et al. (1962). *The myers-briggs type indicator*, volume 34. Consulting Psychologists Press Palo Alto, CA.

National Youth Council of Ireland (2025). Nyci red c cost of living impact study 2025. https://www.youth.ie/documents/nyci-red-c-cost-of-living-impact-study-2025/. Accessed: 2026-03-11.

Nauta, M. M. (2010). The development, evolution, and status of holland's theory of vocational personalities: Reflections and future directions for counseling psychology. *Journal of counseling psychology*, 57(1):11.

Newport, C. (2016). *So good they can't ignore you*. Hachette UK.

Nye, C. D., Su, R., Rounds, J., and Drasgow, F. (2012). Vocational interests and performance: A quantitative summary of over 60 years of research. *Perspectives on Psychological Science*, 7(4):384–403.

O'Connor, R. (2025). Cost of living guide - tu dublin 2025/26. https://www.tudublin.ie/media/website/for-students/student-services-and-support/accommodation/documents/Cost-of-Living-Guide-TU-Dublin-2025.26-.pdf. Accessed: 2026-03-11.

O'Keefe, P. A., Dweck, C. S., and Walton, G. M. (2018). Implicit theories of interest: Finding your passion or developing it? *Psychological science*, 29(10):1653–1664.

Qamhieh, M., Sammaneh, H., and Demaidi, M. N. (2020). Pcrs: personalized career-path recommender system for engineering students. *IEEE Access*, 8:214039–214049.

Roese, N. J. and Summerville, A. (2005). What we regret most... and why. *Personality and Social Psychology Bulletin*, 31(9):1273–1285.

Schafer, J. B., Konstan, J., and Riedl, J. (1999). Recommender systems in e-commerce. In *Proceedings of the 1st ACM conference on Electronic commerce*, pages 158–166.

Slattery, O., Prendergast, M., and Riordáin, M. N. (2023). Navigating a male dominated domain: Experiences of female stem students in higher education in ireland. *Irish Educational Studies*, 42(4):861–880.

Smyth, E., Banks, J., and Calvert, E. (2011). From leaving certificate to leaving school. *The Economic and Social Research Institute*.

Spokane, A. R. and Cruza-Guet, M. C. (2005). Holland's theory of vocational personalities in work environments. *Career development and counseling: Putting theory and research to work*, 24.

Super, D. E. (1980). A life-span, life-space approach to career development. *Journal of vocational behavior*, 16(3):282–298.

Telli Yamamoto, G. (2006). University evaluation-selection: a turkish case. *International Journal of Educational Management*, 20(7):559–569.

Toffler, A. (1970). *Future shock*. Bantam.

Ward, S. (2024). Choosing money over meaningful work: Examining relative job preferences for high compensation versus meaningful work. *Personality and Social Psychology Bulletin*, 50(7):1128–1148.

Xie, Y. and Goyette, K. (2003). Social mobility and the educational choices of asian americans. *Social Science Research*, 32(3):467–498.